JWM Solution Architecture

Deliverables

(Phase 1 / 1)

Discovery Phase Requirements Mapping

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1. Introduction

This document defines the Phase 1 solution architecture deliverables for the JWM Smartsheet Control Center manufacturing production system. Each deliverable is mapped to one or more Requirements (RIDs) identified during the discovery phase conducted between Matt Lynn (Archer Consulting) and Paul Roberts (JWM) across multiple working sessions.

The solution replaces the current Epicor-based production tracking with a Smartsheet-native platform that provides scalable project management, shop floor visibility, inventory tracking, and quality management. A custom user interface extends Smartsheet capabilities for manufacturing-specific workflows.

Phase 1 focuses on operational production visibility and management. Automated inventory logic, financial integration, and advanced resource management are identified as future phases

2. Solution Design Qualification

Final solution design is a concurrent process alongside the build and testing phases. Technical approaches to various deliverables and requirements may be adjusted as the solution takes shape and real-world constraints are encountered. While it is not the intention, it is possible that system limitations within Smartsheet, Control Center, or supporting platforms could require a choice between certain requirements or deliverables based on the technical approach needed to resolve another, or limitations associated with that approach. All requirements and deliverables documented herein are targets to achieve, and any adjustments will be communicated and agreed upon with stakeholders before implementation.

3. Approval Basis

This section defines the approval framework for Phase 1 deliverables. Each deliverable and sub-deliverable is categorized to set clear expectations for scope, flexibility, and change management.

3.1 Approved — Locked for Phase 1 Build

The following deliverables and requirements are fully scoped, discussed in detail during discovery, and locked for the Phase 1 build. These represent firm commitments.

- D-01: Smartsheet Control Center Multi-Tier Project Structure (architecture, hierarchy, horizontal sheet layout, blueprint configuration, archiving)
- D-02a: Core Custom UI — Order/Traveler Building and Management (admin/planner tool for building and editing orders via API to Smartsheet)
- D-03: Baseline vs. Revised Date Tracking and On-Time Delivery Metrics
- D-04: Master Customer List and Standardized Nomenclature
- D-05: Shop Floor Collaboration Layer — Work Apps, Reports, and Views
- D-08: Job Overrun and RMA/Rework Management
- D-09: Portfolio-Level Resource and Schedule Configuration (static capacity values)

3.2 Conceptually Approved — Subject to Detailed Design Refinement

The following items are agreed upon in intent and outcome, but the exact technical approach will be validated and potentially adjusted during the design and build phase. Changes to approach do not require a change order as long as the agreed outcome is achieved.

- D-02b: Schedule Calculation Engine — shift management, duration calculations, propagation logic (method may use UI-side calculations, Smartsheet formulas, scripting, or a combination)
- D-02c: Printed Traveler / Shop Document Output — PDF layout, barcode placeholder (document design to be finalized with stakeholder input)
- D-06a: Inventory Visibility and Manual Transactions — project-level and company-level ledger structure, journal entries, allocation (Data Shuttle/Data Mesh vs. scripting for portfolio rollup TBD)
- D-06b: Purchasing, Receiving, and Stock Job Workflow — material procurement and stock job process design subject to detailed workflow mapping
- D-07a: Scrap Tracking and Material Variance — threshold values and trigger rules are configurable defaults, not hard-coded
- D-07b: Non-Conformance Tracking (NCR/SNCR/CAR) — tracking system structure to be finalized with incoming quality manager
- D-10: Dashboards, Reporting, and KPI Framework — general metric categories defined; specific calculations, rollup views, and dashboard layouts require executive stakeholder input

- D-11: Training, Documentation, and Handoff — scope of training materials and admin handoff to be defined closer to delivery
- Data propagation methods across the system (formulas, Data Mesh, scripting, or combination — to be determined per deliverable during build)

3.3 Excluded — Phase 2 / Future Scope

The following items were discussed during discovery but are explicitly excluded from Phase 1. They may be considered for future phases. See Section 8 for the full exclusions list.

- Automated inventory movement, consumption logic, and back-flushing (RID-15, RID-18)
- System-driven quantity adjustments and automated reorder triggers
- Inventory-specific custom UIs and back-end services
- Financial data integration, cost accounting, or budget tracking in Smartsheet
- Full production roster and dynamic resource management
- Customer portal or external-facing collaboration interfaces
- Automated integration with ERP (Epicor) or accounting systems (Spectrum/QuickBooks)

3.4 Change Order Required

Any requirements, deliverables, or scope additions not documented in this deliverable set that are requested after formal approval will go through a change order process. This includes but is not limited to:

- New requirements not identified during discovery
- Expanded scope on existing deliverables beyond what is described in this document
- Requests for automated processes explicitly deferred to Phase 2
- Additional dashboard metrics requiring new data capture not planned in the current architecture
- Integration with external systems not listed in the current scope

4. Platform Risks and Design Considerations

The following platform-sensitive areas represent known design risks tied to the technical choices in this solution. These are surfaced for stakeholder awareness and will be actively managed during the build phase.

- Row Duplication Across Horizontal Department Sheets — Parts and assemblies appearing as rows in multiple work center sheets means data synchronization is critical. Changes to one instance must propagate reliably to all others.
- Synchronization and State Consistency Between Sheets and Custom UI — The custom UI writes to Smartsheet via API through a replicated database layer. Latency, write conflicts, or API failures could create inconsistent states visible to shop floor users in Work Apps.
- Revision Propagation After Release — When parts, routing, or quantities change on an already-released order, the system must update affected rows across all department sheets without corrupting in-progress or completed operations.
- Performance and Administrative Overhead at Projected Scale — 100-200 active POs, 10-20K active items, rows duplicated across department sheets, cross-sheet formulas, and Data Shuttle processes all push Smartsheet platform limits. Performance at full production volume will be validated during the build phase.
- Inventory Accuracy in a Manual Phase 1 Design — Manual journal entries and operator-reported usage inherently introduce lag and human error. The system can flag discrepancies but cannot prevent them without automated processes (deferred to Phase 2).
- Summary and Dashboard Integrity Across Multi-Project Data — Rollup dashboards depend on consistent data structure across all project toolkits. If sheets diverge from template standards due to manual edits or provisioning errors, dashboard formulas and reports may produce incorrect results.

5. Data Ownership and Synchronization Model

Smartsheet is the system of record for all production data in this solution, including orders, parts and assemblies, operation statuses, inventory transactions, and dates. All reporting, dashboards, and Work Apps read directly from Smartsheet data.

The custom UI reads from and writes to Smartsheet via API, but does so through a hosted and replicated database that synchronizes with Smartsheet rather than querying live Smartsheet data directly. This provides performance isolation, caching, and a layer of protection against direct manipulation of the sheet structure by the UI layer.

When the custom UI writes a change — such as a new order, status update, or date modification — that change propagates from the replicated database to Smartsheet and then across the applicable horizontal department sheets. The exact synchronization mechanisms (API calls, Data Shuttle, Data Mesh, scripting, or formula-based) will be determined per data type during detailed design.

Duplicate-row synchronization across department sheets is a core design challenge. Since parts and assemblies exist as rows in multiple work center sheets, the synchronization model must define how updates to one instance flow to all others. This will be addressed during the detailed design phase and is identified as a platform risk area.

Exception handling for sync failures, API write rejections, or data state inconsistencies will be designed into the solution. The approach will be defined during the build phase and documented in operational runbooks delivered as part of D-11.

6. Phase 1 Deliverables

6.1 D-01: Smartsheet Control Center Multi-Tier Project Structure

Design and deploy a scalable Smartsheet Control Center solution that automates the creation of project workspaces for each purchase order or sales order. The structure is multi-tier in functionality and context, organizing the hierarchy of customers and clients, orders (purchase orders and sales orders containing line items and releases), parts, assemblies, sub-assemblies, and operations. However, it may not use a true multi-tier blueprint structure as Smartsheet defines it with parent and child blueprint configurations. If a true multi-tier blueprint is used, Work Apps cannot be attached to child tier projects. The current plan is to deploy three or more single-tier blueprints that function like a multi-tier solution while preserving the ability to use Work Apps at each tier level. Each project workspace (toolkit) includes standardized template sheets for production tracking, inventory, scrap logging, and metadata. Control Center handles provisioning, blueprint copying, profile data injection, and archiving of completed projects. The solution uses a horizontal sheet architecture: wide-format sheets organized by department or work center (flat laser, CNC, press brake, weld, assembly, QC, paint, shipping, etc.) where each row represents a part or assembly that requires work at that station. Rows are duplicated across each applicable department sheet. Dates and quantities feed successionally through the sheets so that each sheet feeds portfolio-level and program-level rollups by project, customer, and work center. This architecture does not use Smartsheet's built-in Gantt, baseline, or dependency features within the production sheets. The solution supports both processing and architectural divisions using a unified architecture with classification-based filtering. Note that processing jobs are fully estimable and schedulable from intake, while architectural jobs may not be fully plannable until engineering completes layouts, sometimes shortly before shop release. The architecture must accommodate both planning models within the same system.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-40	General (Base Scope)	Provide a data location and structure for operational movements of production at scale
RID-41	Design Intent / Overall Deliverable	Phase 1 automates via SCC the creation and scaling of releases, parts, assemblies, and operations
RID-42	Technical Approach to Architecture	SCC multi-tier provisioning, custom UI for admin, Work Apps for shop floor, plus support solutions
RID-43	Multi-Tier Control Center Architecture	PO as project, line items and releases as child projects or sheet sections
RID-44	Customer / Building Parent Tier	Optional parent level for customer or address-based reporting rollup
RID-45	Parts/Assembly/Sub-Assembly Hierarchy	BOM hierarchy represented logically across horizontal department sheets
RID-46	Purchase Order as Top-Level Project	PO is the project container in Control Center
RID-48	Toolkit Sheet Structure	Standardized sheets per project provisioned by Control Center

RID-27	Annual Volume (1M+ parts, 3M jobs)	Architecture handles high volume via scalable project provisioning
RID-28	Active Orders (10-20K items)	Rows distributed across project sheets, not centralized
RID-29	Active Parts (16-20K operational instances)	Each operation is a row distributed across project toolkits
RID-30	Open POs (100-200 active)	Control Center manages 100-200 active projects concurrently
RID-34	Part and Assembly Structure (Processing + Architectural)	Unified structure supports per-part and batch-release tracking
RID-54	Archiving and Year-Over-Year Metrics	Control Center archiving locks data, preserves summary metrics

6.2 D-02: Custom UI for Order and Traveler Management

The custom UI is a web-based tool that serves as the primary interface for administrators, estimators, planners, and schedulers to build, edit, and manage work orders (travelers/routers) within the Smartsheet back end. Access to the custom UI is strictly limited to these administrative and planning roles. Production leadership and the shop floor will not have any access to the custom UI. Shop floor users interact exclusively with production data through role-based Smartsheet Work Apps, reports, and dashboards (see D-05). This separation is a deliberate data protection measure to prevent accidental changes to the underlying data structure.

D-02a: Core Custom UI — Order/Traveler Building and Management

The custom UI presents a manufacturing-specific form experience that understands hierarchical structure: assemblies, sub-assemblies, parts, and operations. Users select from standardized operation codes, assign quantities, materials, and durations, and upon submission the UI writes all necessary rows, statuses, dates, and formulas to the appropriate horizontal department and work center sheets via API. Each part or assembly is placed as a row in every department sheet where it requires work. The UI supports building orders from templates, copying from previous POs, and revision control for parts. The estimate-to-traveler data flow is direct: the BOM built during estimating is effectively the traveler, and the UI facilitates importing or transforming that estimate data into the production sheet structure. Nesting and batching decisions — where parts from multiple releases are combined on the same material run for yield optimization — are supported through the UI's batching and split functions, with visibility into which items were nested together. The UI includes full administrative access for system administrators and scoped role-based access for estimators, planners, and schedulers appropriate to their function.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-11	JWM Specific Custom UI	Core deliverable: custom web-based UI for building and managing orders
RID-42	Estimate-to-Traveler Data Flow	UI facilitates BOM-to-traveler import; supports build-from-template and copy-from-previous-PO

RID-2	Manual Override Capabilities	UI provides controls for admins to override priorities and dates
RID-7	Engineering Approval Gate	Approval workflow triggered when architectural job dates are shortened
RID-8	Batching and Splitting	Split function creates rows across work centers; batching supports nesting visibility
RID-34	Part and Assembly Structure	UI enforces standardized operation codes and hierarchical BOM entry

D-02b: Schedule Calculation and Shift Management

The custom UI performs duration and schedule calculations based on work center capacity data from the portfolio-level configuration sheet (see D-09). When shift schedules change (e.g., welding moves from 8-hour to 12-hour shifts), the UI propagates recalculated durations to all affected operations. Completed operations retain their original schedule data. For in-progress operations, recalculation is based on remaining quantities — for example, if 80 of 100 parts are complete, only the remaining 20 parts are recalculated at the new shift capacity. The exact technical method for schedule calculations and propagation may use UI-side logic, Smartsheet formulas, custom scripting, or a combination, to be determined during the detailed design phase. Note that architectural division jobs may have limited schedulability until engineering completes layouts, unlike processing division jobs which are fully estimable from intake.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-3	Shift Schedule Management	UI performs duration calculations based on work center shift data; recalculates on schedule changes

D-02c: Printed Traveler / Shop Document Output

The custom UI includes a Print Traveler function that generates a PDF version of the traveler/router document for shop floor use. The team agreed that a paper traveler is still required for shop floor operations. However, the printed traveler does not need all the barcodes previously used on the current traveler document, and no barcode functionality is required in Phase 1. Because barcode and QR code scanning is anticipated as a future capability — particularly for inventory and part identification — a placeholder space for a barcode or QR code will be built into the printed traveler document layout so that the feature can be added later without redesigning the document. The final layout and content of the printed traveler will be defined with stakeholder input during the build phase.

This sub-deliverable does not map to specific RTM requirements but supports the parent deliverable.

6.3 D-03: Baseline vs. Revised Date Tracking and On-Time Delivery Metrics

Implement a date tracking system that preserves the original requested or promised ship date as an immutable baseline date column while allowing a separate adjusted or revised date column for scheduling purposes. The requested date is set during project intake and locked to

non-admin users; it serves as the benchmark for on-time delivery calculations. A separate ship target date is used by schedulers and can be modified freely. On-time delivery is calculated by comparing the ready-to-ship date against the original requested date. Importantly, on-time delivery is measured against when JWM is ready to ship, not when the customer actually picks up — if an order is ready on March 13 but the customer does not pick up until April 3, JWM is not penalized. Smartsheet's built-in Gantt baseline feature is not used in the horizontal sheet architecture. Instead, custom date summary sheets track baseline vs. actual for individual assemblies and releases using dedicated date columns and summary-level formulas. **Smartsheet's native baseline only operates at the project level (entire PO or nothing)**, so release-level and assembly-level variance requires the custom approach. Portfolio-level rollup dashboards report on-time delivery by division, by customer, and overall. The exact methods for populating, sharing, and propagating dates and dependency dates throughout the system are not yet finalized. The solution may use cross-sheet formulas and references, custom scripting or coding, Data Mesh, or other native Smartsheet functions. The approach will be determined during the design and build phase based on system performance, scale constraints, and resource limits.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-1	Baseline vs. Revised Date Tracking	Dual-date column system with immutable baseline and adjustable schedule date
RID-2	Manual Override Capabilities	Admins can override revised dates via the custom UI
RID-12	Metrics and KPIs	On-time delivery metrics by customer, division, and overall derived from this date system

6.4 D-04: Master Customer List and Standardized Nomenclature

Create and maintain a master customer list sheet that serves as the single source of truth for all customer names, codes, and identifiers across the entire solution. The sheet is admin-controlled with restricted edit access; all intake forms and UI entry points use dynamic dropdown lists fed from this master sheet so that end users cannot create ad-hoc customer names. This prevents data inconsistency such as the same customer spelled multiple ways. The master list supports strategic client identification for targeted reporting while still accommodating one-off customers. The same standardized-nomenclature approach extends to part numbers and other key reference data to ensure system-wide consistency.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-55	Master Customer List	Single source of truth for customer names; feeds dropdowns across all forms and UI

6.5 D-05: Shop Floor Collaboration Layer — Work Apps, Reports, and Views

Design and deploy the shop floor user experience using Smartsheet Work Apps as the primary collaboration layer. Work Apps provide a role-based, branded portal where each department (welding, flat laser, CNC, assembly, QC, shipping, etc.) sees only the pages, reports, and forms relevant to their function. This is the sole interface for shop floor users, production leadership, and daily operational users. Shop floor users do not access the custom UI or edit raw Smartsheet sheets; they interact through filtered reports, card and board views, and forms that allow status updates and quantity logging. Column locking and role-based permissions prevent unauthorized changes to dates, predecessors, or structural data. Dynamic views may supplement Work Apps for specific use cases. The solution replaces the need for users to navigate Smartsheet directly, presenting a streamlined app-like experience.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-4	Post-Release Data Editing	Restricted editing via Work Apps roles and admin-only UI access
RID-5	Permissions-Based Editing	Work Apps role-based access controls who can view and edit what
RID-9	Custom Note/Attachment per Operation	Every operation row has a notes column visible in reports, card views, and Work Apps
RID-52	Outsourced Operations Visibility	Filtered reports show outsource status, expected return dates, and current location
RID-53	Quality and Shipping Operations	QC and shipping are standard operations with dedicated Work App pages

6.6 D-06: Phase 1 Inventory Visibility and Manual Transaction Framework

This deliverable provides the data structure, entry points, and manual transaction processes for inventory tracking in Phase 1. It is important to understand that this is an inventory visibility and manual transaction framework, not a fully automated inventory control system. Phase 1 does not include perpetual automated inventory movement, a real-time reservation engine, automatic material depletion or consumption logic, or automated replenishment or reorder triggers. Min/max columns are included in the sheet structure as placeholders for future automation, but no active triggers are built in Phase 1. Manual transaction discipline from operators and administrators is required for data accuracy — the system provides the structure and entry points, but accuracy depends on consistent use by personnel. The intended approach for aggregating project-level inventory data to the company-wide master ledger is Data Shuttle or Data Mesh, but this is pending final scoping and confirmation of required calculations. The exact method will be determined during the build phase.

D-06a: Inventory Visibility and Manual Transactions

Build an inventory framework consisting of two levels: project-level inventory sheets within each toolkit and a portfolio-level company-wide inventory ledger. Each project toolkit includes an inventory sheet tracking materials allocated, consumed, and remaining for that order. The company-wide inventory ledger aggregates quantities across all projects. Inventory items track attributes including part number, description, quantity on hand, quantity allocated, status, source (vendor, stock, or customer-provided), and location. The system separately tracks three

inventory categories: JWM stock inventory, order-specific inventory, and customer-provided inventory, with rules preventing cross-contamination (e.g., customer-provided material cannot be used on a different customer's job). Allocation codes designate which department, customer, or project owns a portion of stock. The system supports manual journal entries for cycle count corrections, transfers between projects, and reconciliation adjustments. Materials purchased for a specific job are flagged as reserved and excluded from available-for-use quantities.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-6	Inventory Auditing	Admin function to flag discrepancies, make journal entry corrections, and reconcile stock counts
RID-38	Inventory as Critical Aspect	Phase 1: manual inventory sheets at project and company level
RID-14	Separation of Financial and Operational Inventory	Tracks operational inventory only; financial data stays in Spectrum
RID-19	Journal Entries for Inventory	Manual plus/minus entries at project level; rollup to company ledger
RID-24	Inventory Attributes	Each inventory row tracks cost, weight, status, delivery date, job assignment, location
RID-25	Allocation Coding	Allocation codes designate department, customer, or project ownership of stock portions

D-06b: Purchasing, Receiving, and Stock Job Workflow

Job-specific procurement is tracked with allocation codes identifying the ordering purpose (job number, stock, or project). Received items are flagged by source type: vendor-purchased, stock-pulled, or customer-provided. The stock job workflow supports sub-components produced in independent batches that are placed into inventory and subsequently consumed as raw materials by top-level assemblies. Stock jobs follow the same production tracking process as standard jobs but their output goes to inventory rather than shipping.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-17	Purchasing and Receiving	Job-specific procurement tracked with allocation codes; received items flagged by source
RID-16	Stock Job Workflow	Stock items produced in batches go to inventory; consumed by top-level assemblies

6.7 D-07: Quality — Scrap Tracking and Non-Conformance Management

This deliverable covers the quality-related tracking systems for scrap, material variance, and non-conformance management.

D-07a: Scrap Tracking and Material Variance

Implement a scrap and variance tracking system within each project toolkit. Operators log scrap events via intake forms specifying part number, quantity scrapped, operation, and reason. The system accumulates scrap quantities per operation per order and compares against configurable thresholds. Default thresholds are 10% of order quantity or 25 pieces, whichever is fewer, but these thresholds can be adjusted or overridden per order, per material, per part, or at other levels by an administrator. They are not hard-coded. When thresholds are exceeded, an automated notification is sent to the quality department for investigation; the system does not auto-generate a non-conformance record. Quality personnel can mark investigated items as addressed, which resets the accumulation counter. Scrap data rolls up to quality dashboards showing trends by operation, by part, and by project. Material variance (planned vs. actual sheets or material used) is tracked through the daily reporting and journal entry process.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-20	Scrap and Material Variance	Operators log extra material usage; system flags variance against planned quantities
RID-35	Scrap and Variance (Operator Reporting)	Operators manually report actual quantities used; variances tracked and accumulated

D-07b: Non-Conformance Tracking — NCR, SNCR, and CAR

A tracking system for quality non-conformances and corrective actions. The system handles internal NCRs (generated against JWM production processes), supplier NCRs (SNCRs, generated internally for distribution to vendors and suppliers), and external NCRs (generated by customers against JWM). Corrective Action Reports (CARs) are tracked with status, owner, due date, and resolution. Non-conformance records are tied to specific jobs, part numbers, and operations for dashboard visibility. Configurable trigger rules and escalation thresholds apply to this system as well — defaults are provided but can be adjusted by administrators.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-51	Non-Conformances, NCRs, SNCRs, CARs	Tracking system for internal, supplier, and customer-issued NCRs with corrective action tracking

6.8 D-08: Job Overrun and RMA/Rework Management

Build processes for handling production overruns and return merchandise authorizations. For job overruns, operators submit a form logging extra parts produced with part number, quantity, and comments. An admin reviews and manually allocates overrun parts to future orders, customer purchase, or stock inventory via a holding queue. For RMAs, a dedicated tracking system allows submission of RMA requests against specific POs or jobs with a unique RMA ID.

RMA dispositions include: credit only (no rework — credit is handled in Spectrum, not Smartsheet, but the disposition is recorded), rework at JWM, rework by customer, or remake and reship. RMAs that require rework or remake flow back through the standard production process flagged as RMA work with their own operation tracking. Recut vs. rework decisions (rework consumes labor only; recut requires new material) route through production leadership approval before execution. RMA dashboards provide visibility into quality escapes, disposition types, and trends.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-21	Job Overrun Handling	Form submission for extra parts; admin queue for allocation
RID-22	RMA Process	Separate RMA tracking with unique IDs, dispositions, and production flow integration
RID-23	Recut vs. Rework	Disposition decision tree routing through approval before execution

6.9 D-09: Portfolio-Level Resource and Schedule Configuration

Create a portfolio-level configuration sheet that defines work center capacity, shift schedules, and resource availability. Each work center (flat laser, CNC, press brake, weld, assembly, QC, paint, etc.) has a row specifying hours per day, days per week, and total weekly capacity. In Phase 1, these are static values manually maintained by administrators. The custom UI reads these values when calculating operation durations. When a shift schedule changes, the configuration sheet is updated and the UI propagates recalculated durations to affected operations based on remaining quantities — for example, if 80 of 100 parts are complete, only the remaining 20 parts are recalculated at the new shift capacity. Completed operations are not modified. This quantity-based recalculation means the time impact depends on how many parts remain, not a proportional time reduction. Note that these static capacity values apply uniformly; architectural division jobs may have limited schedulability until engineering completes layouts, and the system must accommodate both predictable processing schedules and less predictable architectural timelines. The system is designed so that Phase 2 could replace static values with calculated values from a production roster without breaking existing functionality.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-3	Shift Schedule Management	Portfolio-level configuration for shift hours and capacity per work center

6.10 D-10: Dashboards, Reporting, and KPI Framework

Design and deploy a multi-level dashboard and reporting framework. The portfolio-level executive dashboard shows overall production health: on-time delivery rates, active project counts, scrap rates, RMA trends, and work center utilization. Division-level dashboards separate processing and architectural metrics using the same underlying data structure; division-level metrics must be volume-weighted and cannot be simple averages due to differences in part

counts and quantities across divisions. Customer-level dashboards (via the optional parent tier or filtered reports) show performance metrics for strategic accounts. Operational dashboards per work center show current work queue, upcoming scheduled items, and status distributions. Quality dashboards aggregate NCR/SNCR/CAR data, scrap trends, and corrective action status. All dashboards leverage Control Center summary sheet data, cross-sheet formulas, and reports. General categories of metrics have been discussed during discovery (on-time delivery, scrap rates, work center utilization, quality trends), but specific metrics, calculations, and rollup views per destination have not yet been provided in detail by stakeholders. The dashboards and reports will be built to surface data from the underlying system structure, but the ability to deliver specific future metric requests is dependent on the foundational data captured during Phase 1. Metrics not anticipated during the design phase may require additional data points, columns, or structural changes that are outside the current scope. Stakeholder input on desired KPIs — particularly from executive leadership (George, Chris) and the incoming quality manager — remains an open action item.

Requirements Satisfied

RID #	Requirement Name	Coverage Notes
RID-12	Metrics and KPIs	Framework for production, quality, and delivery metrics
RID-54	Archiving and Year-Over-Year Reporting	Archived project data preserved for historical trend analysis

6.11 D-11: Training, Documentation, and Handoff

Deliver training materials, operational documentation, and a structured handoff to JWM administrative users at the conclusion of Phase 1. Training will cover system administration of the custom UI, Control Center provisioning and management, Work Apps configuration, inventory transaction procedures, and quality tracking workflows. Paul Roberts will be trained for admin purposes as the primary internal system administrator. Operational runbooks will document standard procedures, exception handling, and escalation paths. The scope of training materials and the handoff process will be defined in detail closer to the delivery date as the final solution design solidifies.

7. Phase 2 / Future Scope Exclusions

- Automated inventory movement, consumption logic, and back-flushing (RID-15, RID-18)
- System-driven quantity adjustments and automated reorder triggers
- Inventory-specific custom user interfaces and back-end services
- Financial data integration, cost accounting, or budget tracking in Smartsheet (financials remain in Spectrum)
- Automated back-flushing of raw materials based on finished part counts
- Min/max reorder automation (columns included in Phase 1 but triggers are not active)
- Full production roster and dynamic resource management (Phase 1 uses static capacity values)
- Customer portal or external-facing collaboration interfaces
- Automated integration with ERP (Epicor) or accounting systems (Spectrum/QuickBooks)

8. RID Cross-Reference Index

The following table provides a complete cross-reference of all RIDs discussed during discovery mapped to their assigned deliverable.

RID	Requirement	Deliverable	Deliverable Name
RID-1	Baseline vs. Revised Date Tracking	**D-03**	Baseline vs. Revised Date Tracking and On-Time Delivery Metrics
RID-2	Manual Override Capabilities	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-2	Manual Override Capabilities	**D-03**	Baseline vs. Revised Date Tracking and On-Time Delivery Metrics
RID-3	Shift Schedule Management	**D-02b**	Schedule Calculation and Shift Management
RID-3	Shift Schedule Management	**D-09**	Portfolio-Level Resource and Schedule Configuration
RID-4	Post-Release Data Editing	**D-05**	Shop Floor Collaboration Layer — Work Apps, Reports, and Views
RID-5	Permissions-Based Editing	**D-05**	Shop Floor Collaboration Layer — Work Apps, Reports, and Views
RID-6	Inventory Auditing	**D-06a**	Inventory Visibility and Manual Transactions
RID-7	Engineering Approval Gate	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-8	Batching and Splitting	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-9	Custom Note/Attachment per Operation	**D-05**	Shop Floor Collaboration Layer — Work Apps, Reports, and Views
RID-11	JWM Specific Custom UI	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-12	Metrics and KPIs	**D-03**	Baseline vs. Revised Date Tracking and On-Time Delivery Metrics
RID-12	Metrics and KPIs	**D-10**	Dashboards, Reporting, and KPI Framework
RID-14	Separation of Financial and Operational Inventory	**D-06a**	Inventory Visibility and Manual Transactions
RID-16	Stock Job Workflow	**D-06b**	Purchasing, Receiving, and Stock Job Workflow
RID-17	Purchasing and Receiving	**D-06b**	Purchasing, Receiving, and Stock Job Workflow
RID-19	Journal Entries for Inventory	**D-06a**	Inventory Visibility and Manual Transactions
RID-20	Scrap and Material Variance	**D-07a**	Scrap Tracking and Material Variance
RID-21	Job Overrun Handling	**D-08**	Job Overrun and RMA/Rework Management

RID-22	RMA Process	**D-08**	Job Overrun and RMA/Rework Management
RID-23	Recut vs. Rework	**D-08**	Job Overrun and RMA/Rework Management
RID-24	Inventory Attributes	**D-06a**	Inventory Visibility and Manual Transactions
RID-25	Allocation Coding	**D-06a**	Inventory Visibility and Manual Transactions
RID-27	Annual Volume (1M+ parts, 3M jobs)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-28	Active Orders (10-20K items)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-29	Active Parts (16-20K operational instances)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-30	Open POs (100-200 active)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-34	Part and Assembly Structure (Processing + Architectural)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-34	Part and Assembly Structure	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-35	Scrap and Variance (Operator Reporting)	**D-07a**	Scrap Tracking and Material Variance
RID-38	Inventory as Critical Aspect	**D-06a**	Inventory Visibility and Manual Transactions
RID-40	General (Base Scope)	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-41	Design Intent / Overall Deliverable	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-42	Technical Approach to Architecture	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-42	Estimate-to-Traveler Data Flow	**D-02a**	Core Custom UI — Order/Traveler Building and Management
RID-43	Multi-Tier Control Center Architecture	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-44	Customer / Building Parent Tier	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-45	Parts/Assembly/Sub-Assembly Hierarchy	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-46	Purchase Order as Top-Level Project	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-48	Toolkit Sheet Structure	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-51	Non-Conformances, NCRs, SNCRs, CARs	**D-07b**	Non-Conformance Tracking — NCR, SNCR, and CAR
RID-52	Outsourced Operations Visibility	**D-05**	Shop Floor Collaboration Layer — Work Apps, Reports, and Views

RID-53	Quality and Shipping Operations	**D-05**	Shop Floor Collaboration Layer — Work Apps, Reports, and Views
RID-54	Archiving and Year-Over-Year Metrics	**D-01**	Smartsheet Control Center Multi-Tier Project Structure
RID-54	Archiving and Year-Over-Year Reporting	**D-10**	Dashboards, Reporting, and KPI Framework
RID-55	Master Customer List	**D-04**	Master Customer List and Standardized Nomenclature